

SHIVAM RAWAT

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Summary

Software engineer with experience building and maintaining a production academic assessment platform at IIT Bombay, contributing across backend, web, Android, and iOS modules. Worked on authentication flows, attendance integrity mechanisms, asynchronous processing, and an ML-powered grading pipeline in live examination environments. Comfortable debugging large codebases and implementing scalable, reliable logic across systems.

Experience

SAFE — IIT Bombay | Senior Project Technical Assistant

05/02/24 – Present

- Built an automated answer-sheet grading module that extracts handwritten roll numbers from scanned exam PDFs using a TrOCR (PyTorch) OCR pipeline with OpenCV preprocessing (deskew, contrast enhancement, upscaling), auto-crops per-question answer images, and exports a graded-submissions CSV — replacing instructors' manual sorting workflow.
- Offloaded the ~1.3 GB OCR model and image-extraction work onto dedicated, isolated Celery queues separate from the web and proctoring workers, and built an async polling job system with deadline-based recovery for stuck jobs; diagnosed and fixed a native tokenizer segfault that was crashing the OCR worker.
- Redesigned attendance proximity verification for iOS after MAC address deprecation by implementing a custom BLE scanning and advertising mechanism with device identity persistence using iOS Keychain.
- Implemented a submission recovery mechanism using browser local storage and backend reconciliation logic, allowing students to resume interrupted quizzes without data loss.
- Built a scientific calculator and an instructor–student chat for the iOS examination app, supporting in-exam calculations and post-quiz query resolution between students and instructors.
- Diagnosed and resolved production-breaking issues across quiz submission, webquiz, and attendance flows by tracing execution across Django views, Celery workers, and frontend state; stabilized asynchronous submission processing by fixing data-type inconsistencies and adding a locking mechanism to prevent concurrent Celery task execution.

Skills

Languages: Python, C++, JavaScript

Backend: Django, Flask, REST APIs, SQL, Celery, asynchronous task processing

ML / CV: PyTorch, Hugging Face Transformers (TrOCR), OpenCV, OCR

Frontend: Vue.js, React

Mobile: Android (Java), iOS (Swift)

Core CS: Data Structures, Operating Systems, System Design, DBMS

Tools & Platforms: Git, Docker, Linux, Selenium, Android Studio, Xcode, VSCode

Projects

Code Clasher | Django, Django Channels, React

[View code](#)

A real-time 1v1 competitive coding battle arena with ranked matchmaking, live health bars, and in-browser code execution.

- Built real-time 1v1 coding battles where two players race on the same problem with a live health/damage system, using Django Channels over WebSockets (Redis channel layer) to sync match state, health, and results across clients.
- Implemented JWT authentication, Elo-based ranked matchmaking across skill tiers, and a sandboxed Python execution engine that runs user submissions against test cases under per-problem time limits.
- Configured the Daphne/ASGI Django backend and React + Vite client for production with environment-configurable API/WebSocket endpoints and CORS.

MockMate | Next.js, Django, Gemini, Deepgram

[View code](#)

An AI voice interviewer that analyzes a resume, holds a real-time spoken interview, and returns a scored feedback report.

- Built a real-time voice interview where a Gemini-driven interviewer speaks each question (Deepgram Aura text-to-speech) and the candidate answers by voice, with automatic turn-taking from in-browser silence detection — answers are recorded, uploaded, and transcribed server-side (Deepgram).
- Modeled the interviewer as a stateful Gemini conversation over the full transcript (asks follow-ups, repeats or clarifies on request, and wraps up), using Gemini structured outputs (validated Pydantic schemas) for resume parsing, question generation, and the final scored report.
- Deployed a Django REST + JWT backend and a Next.js (App Router) frontend on Vercel with PostgreSQL, proxying all Deepgram traffic through the backend to keep the master API key server-side.

Education

Hemwati Nandan Bahuguna Garhwal University, Srinagar, Uttarakhand

2019 – 2023

- B.Tech, Computer Science & Engineering